MISO System Load Analysis

Growth, Patterns, and Peak Demand (2023-2025)

Thom Johnson, MS
Business Analytics

Author's Note

This analysis was conducted following my completion of the Master of Science in Business Analytics program at the University of Iowa in December 2025. The project was undertaken as a self-directed extension of my academic work, with the goal of developing a rigorous, portfolio-quality analysis using real-world data from a critical infrastructure system.

During graduate school, I served as a teaching assistant for undergraduate macroeconomics courses. That experience reinforced the central role of the utility sector—and electricity in particular—in supporting modern economic activity. Reliable electric power underpins virtually every dimension of contemporary life, from basic household needs to industrial production, financial systems, healthcare, and public safety. Disruptions to the electric system, even over short periods, can have outsized and cascading effects.

With this context in mind, I sought out high-quality, publicly available datasets that would allow for meaningful exploration of system behavior at scale. MISO's Actual Load data provided a compelling opportunity. Analyzing load dynamics across a large portion of the U.S. grid offered a way to deepen my understanding of electric system operations while applying analytics techniques to a dataset with direct relevance to system planning, reliability, and market participants.

This analysis reflects both a professional interest in the utility industry and a broader appreciation for the importance of electrical infrastructure. It is intended as a technical exploration of observed system behavior and as a demonstration of applied analytics in a real-world, high-stakes domain.

Executive Summary

Over the coming years, the U.S. power grid will face levels of demand that challenge it in ways not previously experienced. Electricity demand across large U.S. power systems is entering a period of sustained growth and structural change. Broad electrification trends, expanding data center deployment, and economic growth and urbanization are placing upward pressure on baseline system demand, while weather variability and tighter operating margins increase exposure to extreme load conditions. [1]

At the same time, the pace at which new generation, transmission, and supporting infrastructure can be planned, permitted, and interconnected has emerged as an increasingly important constraint on system flexibility. In this environment, understanding not only how high system load can reach, but also how frequently systems operate near their upper limits, has become central to understanding system behavior and the challenges facing the grid.

This analysis examines observed load behavior across the Midcontinent Independent System Operator (MISO) footprint from September 2023 through December 2025 using hourly actual load data. The analysis is intentionally descriptive, focusing on how system-level demand evolves over time, how peak demand conditions emerge and persist, and how those conditions manifest across regions. Particular emphasis is placed on the frequency, structure, and timing of extreme system load.

System demand across MISO is rising, and peak demand conditions are occurring more frequently as a result. Over the analysis period, median hourly system load increased by approximately **8.6 percent**, while total energy consumption rose by roughly **4 percent** from 2024 to 2025. Together, these results indicate both a higher typical operating level and greater aggregate demand, signaling an upward shift in baseline system conditions rather than isolated increases at the upper tail.

Consistent with this trend, peak demand exposure has accelerated sharply in recent years. Peak demand hours, defined as hours at or above the **99th-percentile threshold of 110,423 MW**, are increasingly concentrated in the most recent year, with approximately **77 percent occurring in 2025 alone**. Peak demand days exhibit a similar pattern, increasing by roughly **3.4 times from 2024 to 2025**, with nearly **75 percent of all peak demand days concentrated in 2025**. This pattern reflects more frequent entry into extreme load conditions rather than longer or more severe individual events.

Despite this increase in frequency, peak demand events retain a stable internal structure. Events most commonly last **four consecutive hours**, with no observed events exceeding **ten hours** in duration. This bounded profile indicates that extreme load conditions remain confined within a single operating day, even as their occurrence becomes more common.

Peak demand in MISO is most likely to occur on summer weekday afternoons, particularly in July and August. Nearly all peak demand hours occur during the summer months, with **July and August accounting for approximately 83 percent of peak demand days**. Peak demand is disproportionately concentrated on weekdays, and **87 percent of peak hours occur between 1:00 PM and 7:00 PM Eastern Time**, reflecting a tightly defined seasonal and intraday operating window.

Conditions leading into peak demand further indicate that extreme load outcomes are not driven by **abrupt intraday ramp-up periods**. On peak demand days, morning system load is already elevated by approximately **8,000 to 10,000 MW** relative to non-peak days and follows parallel hourly trajectories throughout the day. Peak demand therefore emerges from persistently high baseline load rather than short-term shocks.

Peak demand also reflects system-wide behavior rather than regional anomalies. While the Central region consistently carries the largest share of system load, regional load shares remain broadly stable during peak demand hours, with only modest and predictable reweighting relative to non-peak conditions. Seasonal load patterns are consistent across regions, reinforcing that peak demand reflects coordinated system-level demand growth rather than localized effects.

Taken together, these findings indicate that MISO is operating at higher load levels more often and entering peak demand conditions at or above **110,423 MW** with increasing frequency during predictable summer windows. Rising peak exposure is driven by sustained growth in baseline demand interacting with shared temporal and seasonal drivers, not by increasingly extreme or prolonged peak events.

As load growth continues, maintaining reliable generation, transmission readiness, and operational flexibility during recurrent peak periods will become increasingly central to system planning and operations. Understanding how peak demand emerges before reliability margins are tested provides a critical foundation for informed resource planning, market design, and long-term investment decisions across the MISO footprint.

Table of Contents

Author's Note 2
Executive Summary 3
Table of Contents 5
Background & Scope 6
Data & Methodology 10
Results & Findings 14
Limitations & Next Steps 37
Works Cited 40

Background & Scope

This analysis examines system load behavior within the Midcontinent Independent System Operator (MISO), the regional transmission organization responsible for maintaining reliability and operating wholesale electricity markets across a large portion of the central North American power grid. As both a market operator and a system operator, MISO plays a critical role in balancing electricity supply and demand in real time, coordinating generation and transmission across a geographically expansive and operationally complex network.

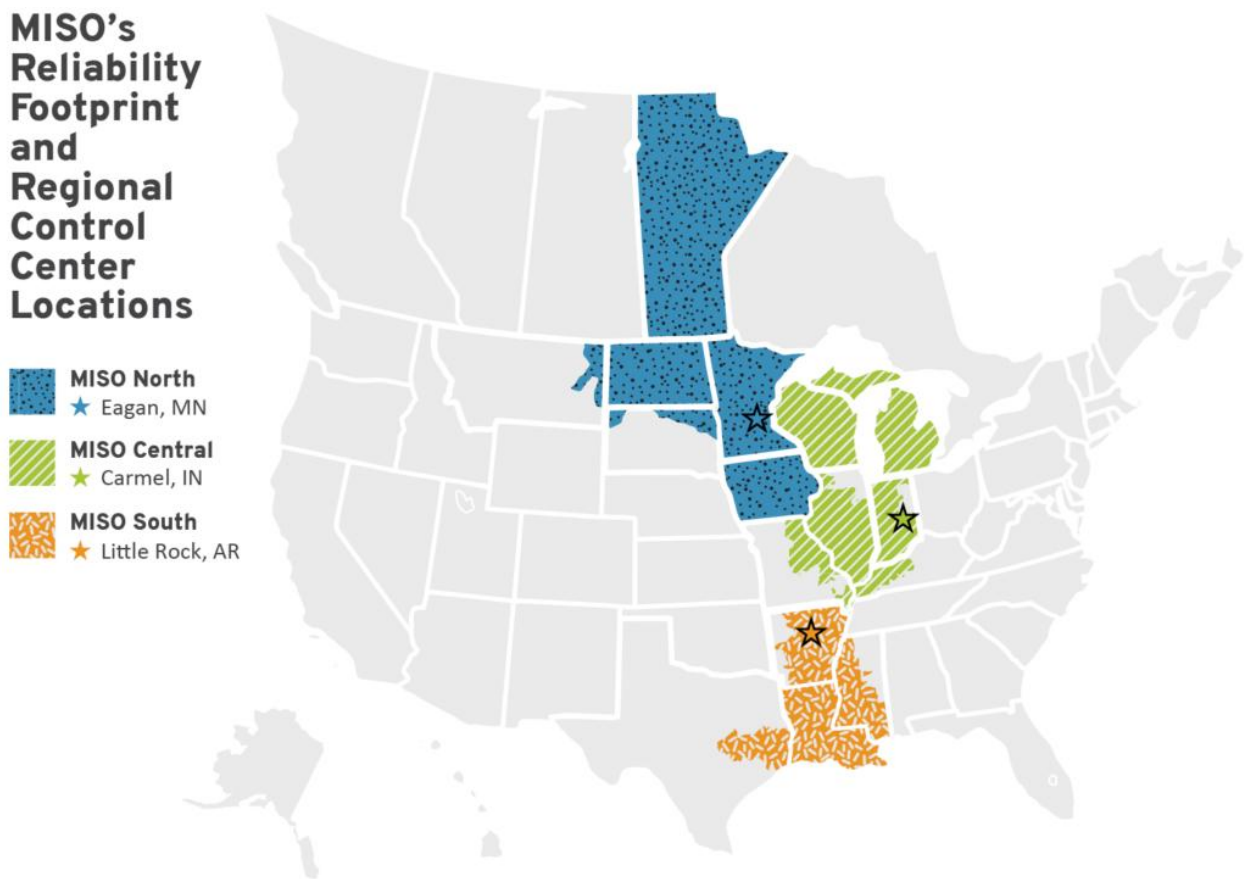

Image source: R Street Institute [2]

System Footprint

MISO's operational footprint spans 15 U.S. states and the Canadian province of Manitoba, serving approximately 45 million people. The system includes a diverse mix of load-serving entities, transmission owners, generators, and market participants, encompassing vertically integrated utilities, independent power producers, municipal utilities, electric cooperatives, and power marketers. [3]

Geographically, MISO covers a wide range of economic activity and climate zones, including major metropolitan centers, industrial corridors, agricultural regions, and both northern and southern temperature regimes. This diversity introduces meaningful variation in load shape, seasonal demand drivers, and peak stress conditions across the system, making MISO a particularly informative case study for understanding large-scale grid behavior.

Operationally, MISO is organized into three regions—North, Central, and South—which reflect differences in climate, load composition, and infrastructure characteristics. These regions are shown on the previous page and are examined in greater detail later in the analysis.

Object of Measurement

The primary object of study in this analysis is system load, representing the total electrical demand served across the Midcontinent Independent System Operator (MISO) footprint. At the system level, load reflects the aggregate outcome of residential, commercial, industrial, and other electricity consumption occurring simultaneously across a large and interconnected grid.

System load is a foundational quantity in power system operations and planning. It defines the level of demand that generation, transmission, and operating reserves must collectively satisfy to maintain real-time balance and reliability. Changes in load shape, magnitude, and timing therefore have direct implications for system stress, peak conditions, and operational decision-making.

In this analysis, load is examined at an hourly cadence, consistent with the temporal resolution used in bulk power system operations and market settlement. This perspective emphasizes sustained operating conditions and recurring demand patterns rather than momentary fluctuations, making it well suited for characterizing both typical system behavior and periods of elevated demand.

This subsection establishes the conceptual meaning and relevance of system load. The technical representation of load in the source data—including measurement conventions, units, and aggregation methods—is described in detail in the *Data & Methodology* section, which specifies how load is operationalized for analysis.

Sources of System Load

Observed system load reflects the aggregate outcome of multiple demand sources operating simultaneously across the MISO footprint:

- **Residential demand** is primarily weather-driven and closely tied to household behavior. Space cooling during summer heat events and electric heating during winter cold snaps produce large, highly coincident increases in demand, often acting as the marginal factor that pushes system load into peak conditions. [4]

- **Commercial demand**, including offices, retail establishments, hospitals, schools, public facilities, and large digital infrastructure such as data centers, generally follows institutional schedules and business-hour patterns. While traditional commercial load exhibits strong diurnal and weekday structure, data center demand operates at much higher load factors and contributes to sustained baseline demand. Together, these uses shape the daily load profile and contribute to the persistence of elevated demand across consecutive daytime hours. [5]
- **Industrial demand** provides a substantial and relatively stable baseline. Energy-intensive activities such as manufacturing, food processing, metals, and chemicals operate at scales that materially influence both regional and system-level load, raising the overall floor of demand and affecting both peak magnitude and duration. In combination with growing digital and electrified industrial processes, this sector contributes to higher sustained load levels even outside traditional peak windows. [6]
- **Agricultural and seasonal demand**, including grain drying, irrigation, and livestock operations, introduces additional regional and seasonal variation, particularly in shoulder seasons. While smaller in aggregate, these loads can amplify regional stress when coincident with other demand drivers. [7]

Peak load conditions in MISO therefore emerge not from any single sector acting in isolation, but from the coincidence of temperature-sensitive residential demand with sustained commercial and industrial activity. Load is best understood as the combined expression of physical systems, economic behavior, and environmental conditions rather than the result of isolated consumption decisions.

Temporal Scope

The analysis covers the period from September 1, 2023 through December 31, 2025, using continuous hourly observations. This timeframe spans multiple seasons, including two summer peak periods, and captures repeated daily, weekly, and seasonal demand cycles. The selected timeframe represents the available MISO actual load data at the time of writing and is intended to be sufficiently long to distinguish persistent structural patterns from short-term anomalies.

All timestamps are in Eastern Standard Time (EST) and reflect authoritative clock time as reported in the source data.

Seasonal Context

Seasonal variation plays a central role in shaping electrical load behavior across the MISO footprint. Differences in temperature, daylight, and economic activity produce recurring seasonal demand patterns that are especially relevant when examining peak load conditions.

For the purposes of this analysis, seasons are defined using standard meteorological groupings consistent with common utility planning practice. These seasonal categories are used to

characterize typical demand behavior and to support interpretable comparisons across time and regions, rather than to define operational thresholds or planning constructs. Formal seasonal definitions and their application are described in detail in the *Data & Methodology* section.

Purpose and Scope of Analysis

The purpose of this study is to develop a clear, system-level understanding of how electrical load behaves within the Midcontinent Independent System Operator (MISO) footprint under both typical and elevated demand conditions. The analysis characterizes the structure, timing, persistence, and regional composition of demand, with particular emphasis on extreme system load.

The study is intentionally descriptive and exploratory. Rather than evaluating reliability outcomes or market performance, it documents how system-level electrical load behaves across time and regions, providing a factual foundation for future operational, planning, or market-focused analyses.

The analysis proceeds from broad system behavior to increasingly granular temporal and regional perspectives, examining:

- **System-level load trends and distribution**, including growth, variability, and the distinction between typical and extreme demand conditions
- **Temporal characteristics of peak demand**, including seasonal, monthly, weekly, and intraday patterns
- **Persistence of peak demand events**, focusing on duration and multi-hour structure
- **System conditions leading into peak demand**, with emphasis on baseline load levels rather than short-term ramps
- **Regional load behavior**, including baseline differences, seasonal variation, and stability of regional load shares during peak conditions

Across all sections, peak demand is defined using a percentile-based approach rather than absolute maxima. Specifically, peak demand is defined as the 99th percentile (p99) of system-level hourly load, calculated globally across all observations in the study period. This definition captures rare but operationally significant demand conditions while limiting sensitivity to isolated outliers.

The scope of this work is limited to observed load behavior. The analysis does not assess system reliability, capacity adequacy, transmission constraints, or market outcomes, as such evaluations require additional supply-side, weather, and operational data beyond the available dataset. Forecasting and predictive modeling are outside the scope of this phase and may be addressed in future work.

Data & Methodology

Data Sources

Load data used in this analysis were obtained from the Midcontinent Independent System Operator (MISO) Data Exchange via its public REST API. The study relies on MISO's Actual Load dataset, which reports realized electrical load used in the real-time market at hourly resolution and is published daily. The dataset includes both total system load and region-level load values, enabling consistent analysis at multiple levels of aggregation from a single authoritative source. [8]

Temporal Scope and Resolution

The analysis covers the period from September 1, 2023 through December 31, 2025, using continuous hourly observations. This window spans multiple seasons and includes two full summer peak periods, enabling the identification of recurring structural patterns rather than isolated anomalies.

All timestamps are reported and analyzed in Eastern Standard Time (EST). No daylight-saving time adjustments are applied in the source data. Each observation corresponds to a clearly defined one-hour operating interval.

Load Definition and Measurement

This subsection describes how load is treated technically within the dataset; conceptual context and system interpretation are provided in the Background & Scope section. In the source dataset, load values are reported in megawatt-hours (MWh) for each operating interval. Because each interval represents exactly one hour, these values are interpreted as the average load in megawatts (MW) over that hour, which is numerically equivalent to MWh per interval. This treatment is consistent with standard ISO reporting conventions and bulk power system analysis practice. Throughout this study, load is therefore expressed and analyzed in units of MW.

System load represents the aggregate electrical demand that must be supplied to maintain real-time balance across the MISO footprint during each operating hour. Observations reflect sustained hourly operating conditions rather than instantaneous within-hour maxima. As a result, the data are well suited for evaluating system-level trends, characterizing peak demand conditions, and assessing the timing and persistence of elevated load states relevant to operational planning and system analysis.

Seasonal Definitions

Seasonal classifications are applied in this analysis to support consistent temporal aggregation and comparison of load behavior across years and regions. Seasons are defined using the standard meteorological groupings employed by MISO in public reporting and commonly used in utility planning and system analysis. [9]

The following definitions are applied uniformly across the full study period and across all MISO regions:

- **Winter:** December–February
- **Spring:** March–May
- **Summer:** June–August
- **Fall:** September–November

Season assignments were created during data ingestion and applied consistently to all hourly observations in the dataset. These definitions are used throughout the analysis to characterize seasonal load patterns, assess the seasonal concentration of peak demand, and compare regional load behavior under common temporal groupings.

By using MISO-consistent seasonal definitions, the analysis maintains alignment with industry practice and ensures that observed seasonal patterns reflect underlying system behavior rather than analyst-defined categorizations.

Peak Demand Identification

Peak demand is defined using a percentile-based approach rather than absolute maxima. Specifically, peak demand is defined as the 99th percentile (p99) of system-level hourly load, computed globally across all observations in the full study period. The resulting threshold of 110,423 MW is held fixed throughout the analysis and applied consistently across years, seasons, and regions.

Using this definition:

- Peak demand hours are hourly intervals with system load at or above the p99 threshold.
- Peak demand days are calendar days containing at least one peak demand hour.
- Peak demand events are defined as uninterrupted sequences of consecutive peak demand hours.

This framework isolates rare but operationally significant demand conditions while avoiding undue sensitivity to isolated outliers.

Statistical Measures and Aggregation

Median-based statistics are used as the primary measures of central tendency throughout the analysis to reduce sensitivity to extreme values in a right-skewed load distribution. Percentiles, including the p99 threshold, are computed exactly from the empirical distribution of observed values.

Long-term system trends are evaluated using a 12-month trailing rolling median, computed from monthly median system load values. This two-stage aggregation smooths short-term volatility while preserving shifts in the underlying level of system demand.

Data Ingestion and Storage

Load data used in this analysis were retrieved programmatically from the Midcontinent Independent System Operator (MISO) Data Exchange using its public REST API. The Actual Load dataset was accessed via authenticated API requests and ingested using Python-based data extraction scripts.

Raw hourly observations were parsed, validated, and stored in a structured format before being loaded into Google BigQuery for analysis. This approach enabled efficient handling of large time-series datasets and supported repeatable, auditable querying across system-level and regional load views derived from the same source data.

Querying and Analysis Environment

All analytical queries were executed using standard SQL in Google BigQuery. The dataset was structured to support both system-level aggregation and region-level analysis using the regional identifiers provided directly in the MISO Actual Load dataset.

SQL was used to compute descriptive statistics, percentiles, rolling medians, event groupings, and temporal aggregations. Median-based measures were favored where appropriate to reduce sensitivity to extreme values in a right-skewed load distribution. All derived metrics used in figures and tables reflect direct query outputs from this environment.

Visualization and Figure Construction

Figures presented in this report were constructed using Microsoft Power BI. Query outputs from BigQuery were used as the data source for visualization, with charts designed manually to emphasize interpretability, consistency, and alignment with the analytical questions addressed in each section.

Power BI was used exclusively for visualization and presentation; all analytical logic and statistical computation were performed upstream in SQL. This separation ensures that figures reflect reproducible analytical results rather than embedded or opaque transformations within the visualization layer.

Use of Large Language Models (LLMs)

Large language models were used in a supporting role during the development of this analysis to assist with code drafting, text revision, and structural refinement. All analytical logic, statistical definitions, interpretations, and final written content were either generated directly or reviewed, validated, and finalized by the author.

The use of LLMs was limited to productivity support and did not replace independent reasoning, verification of results, or author judgment. Responsibility for all conclusions and representations in this report rests solely with the author.

Analytical Scope and Limitations

This study is intentionally descriptive and exploratory, focusing exclusively on observed load behavior. It does not evaluate system reliability, resource adequacy, transmission constraints, market prices, or weather-driven attribution, as such analyses require additional supply-side, meteorological, and operational data beyond the available dataset.

Forecasting, causal modeling, and scenario analysis are outside the scope of this phase and may be addressed in future work building on the empirical patterns documented here.

Results & Findings

This section examines system-level load behavior, the timing and persistence of peak demand conditions, and how those conditions manifest across the MISO footprint. The analysis proceeds from broad system trends to increasingly granular temporal and regional perspectives, addressing the following questions in sequence:

System Load Trends and Distribution

- Is system load increasing?
- Did energy consumption increase from 2024 to 2025?
- How does system load distribute under normal and extreme conditions?
- How distinct are peak demand hours from typical system load?

Temporal Characteristics of Peak Demand

- How are peak demand hours distributed across years?
- How are peak demand days distributed across years?
- Is peak demand seasonally concentrated?
- What months do peak demand days occur?
- Do peak demand days occur more often on weekdays or weekends?
- At what times of day do peak demand hours most often occur?

Persistence and Duration of Peak Demand Events

- How long do peak demand events persist?
- What duration ranges do peak demand events most commonly fall into?

System Conditions Leading into Peak Demand

- What load conditions precede peak demand events?

Regional Load Behavior and Composition

- How do MISO regions differ in demand?
- Is peak demand driven by a shift in regional load shares?
- How does seasonal load behavior differ across MISO regions?

A foundational question for any assessment of system load is whether overall electricity demand is rising over time. Before examining peak behavior, regional dynamics, or extreme demand events, it is essential to first establish whether the underlying level of system load itself is changing.

Is system load increasing?

To evaluate whether system load is rising over time, this analysis examines the **12-month trailing rolling median of system load**, computed from monthly medians derived from hourly

data. This approach smooths short-term volatility and seasonal noise while preserving shifts in the system's central tendency. By relying on medians rather than averages, the metric is robust to extreme weather events and other transient shocks, making it well suited for identifying structural changes in demand.

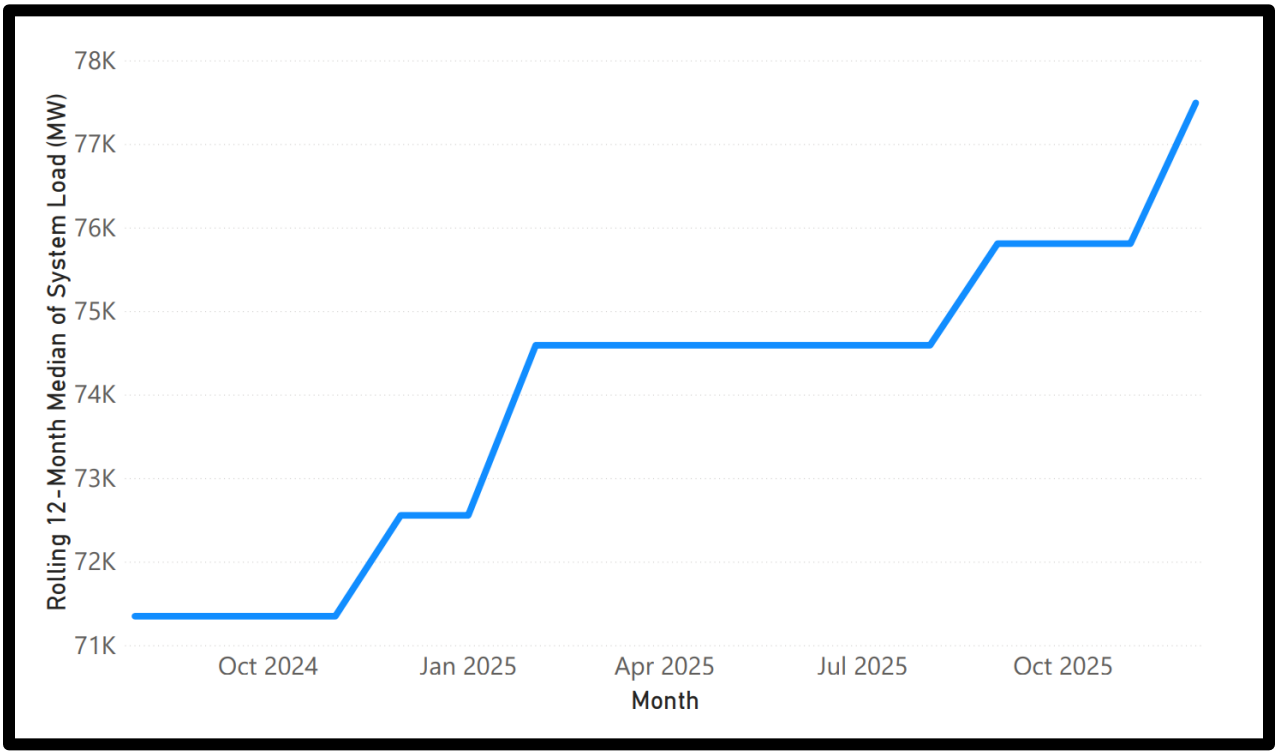

Figure 1.1. Rolling 12-Month Median of System Load (MW)
The figure displays the trailing 12-month median of monthly system load, calculated from hourly observations. Each point represents the median load across the most recent twelve months, providing a stable view of changes in the system's typical demand level over time.

The results show a clear upward shift in median system load. The rolling median begins at **71,348 MW** in the first full observation window and rises to **77,492 MW** by the end of the study period—an increase of **6,144 MW**, or approximately **8.6%**. Rather than increasing smoothly, the series advances in discrete steps, reflecting sustained upward re-ranking of monthly load levels as higher-demand months replace lower-demand months within the rolling window.

This pattern indicates that the observed increase is not driven by isolated spikes or short-lived anomalies, but by a persistent elevation in typical system demand. On this basis, the answer to the central question is unambiguous: **median system load is increasing**.

Based on findings that the baseline level of system load is rising over time, the next question is whether this upward shift reflects a meaningful increase in overall electricity consumption. To

assess this, the analysis turns to **year-over-year changes in total energy use**, comparing aggregate system consumption in **2024 and 2025**.

Did energy consumption increase from 2024 to 2025?

To complement the analysis of median system load, this section examines whether total electricity consumption across the MISO system also increased over time. Whereas median-based metrics capture changes in the typical operating level of demand, total energy consumption reflects the aggregate volume of electricity demanded and provides an additional lens on system growth.

Total system energy consumption is calculated by summing hourly system load across all hours in each calendar year. Using this measure, total energy consumption increased from approximately **653 million MWh in 2024** to **679 million MWh in 2025**, representing an increase of roughly **26 million MWh**, or **about 4% year over year**.

Unlike median load metrics, total energy consumption is more sensitive to weather variability, calendar effects, and short-term fluctuations in demand. Despite this sensitivity, the observed increase in total energy use aligns closely with the upward shift in the rolling 12-month median of system load documented in the previous section. The consistency across these two measures—one focused on typical load levels and the other on aggregate volume—indicates that system growth is not confined to isolated periods or driven solely by extremes.

Taken together, the results establish that rising demand in the MISO system reflects both a **higher baseline level of load** and **greater overall energy consumption**. This dual increase reinforces the conclusion that system load is growing in a broad and persistent manner, rather than through transient or anomalous effects.

With results indicating that both median system load and total energy consumption are increasing, the analysis now turns to the **distribution of hourly load**. This shift enables a closer examination of how often different load levels occur and how the upper tail of the distribution behaves, providing the foundation for defining peak demand conditions and identifying the specific hours that drive system load in subsequent sections.

How does system load distribute under normal and extreme conditions?

Hourly system load across the MISO footprint exhibits a well-defined, right-skewed distribution, with most observations concentrated within a relatively narrow operating range and a long upper tail corresponding to infrequent, high-demand conditions. Examining this distribution provides critical context for distinguishing typical system behavior from the extreme states that define peak demand.

Metric	System Load (MW)
Minimum	52,892
Average	75,225
Median (p50)	73,148
90th percentile (p90)	91,635
95th percentile (p95)	98,868
99th percentile (p99)	110,423
Maximum	121,339

Table 1.1. Hourly System Load Distribution Summary (MW)
Summary statistics of hourly system load across the MISO footprint for the full study period (2023-09-01 to 2025-12-31).

As shown in Table 1.1, typical system load centers in the low-to-mid 70,000 MW range, with a **median of 73,148 MW** and an average of 75,225 MW. The close alignment between the median and mean indicates a stable central tendency in system demand under normal operating conditions, suggesting that typical load levels are not dominated by extreme outliers.

As load increases toward the upper tail of the distribution, dispersion grows more rapidly. The 90th percentile occurs at 91,635 MW and the 95th percentile at 98,868 MW, while the 99th percentile rises sharply to **110,423 MW**. In this analysis, the 99th percentile threshold is used to define peak demand hours. The widening separation between percentile thresholds illustrates the nonlinear escalation of system load as demand approaches extreme conditions, with relatively small increases in percentile rank corresponding to large increases in megawatt magnitude.

Maximum observed load during the study window reaches **121,339 MW**, well above the p99 threshold. For additional context, the historical all-time MISO system peak is approximately 127,000 MW, recorded on July 20, 2011. The proximity of these values reflects the influence of rare, highly coincident weather-driven peak events, as absolute system maxima are episodic and not directly indicative of underlying load growth.

While Table 1.1 characterizes the overall distribution of system load and identifies the upper tail numerically, it does not convey how distinctly peak demand hours separate from typical operating conditions in practice. To assess whether peak demand represents a gradual extension of normal load or a fundamentally different operating regime, the next section examines the full distribution of hourly system load and explicitly isolates the hours that define peak demand.

How distinct are peak demand hours from typical system load?

With the overall distribution of system load and the numerical threshold for extreme conditions defined, this section examines how peak demand hours relate to the broader load profile. Specifically, it assesses whether peak demand hours blend smoothly into the upper end of typical system operations or occupy a clearly distinct region of the load distribution.

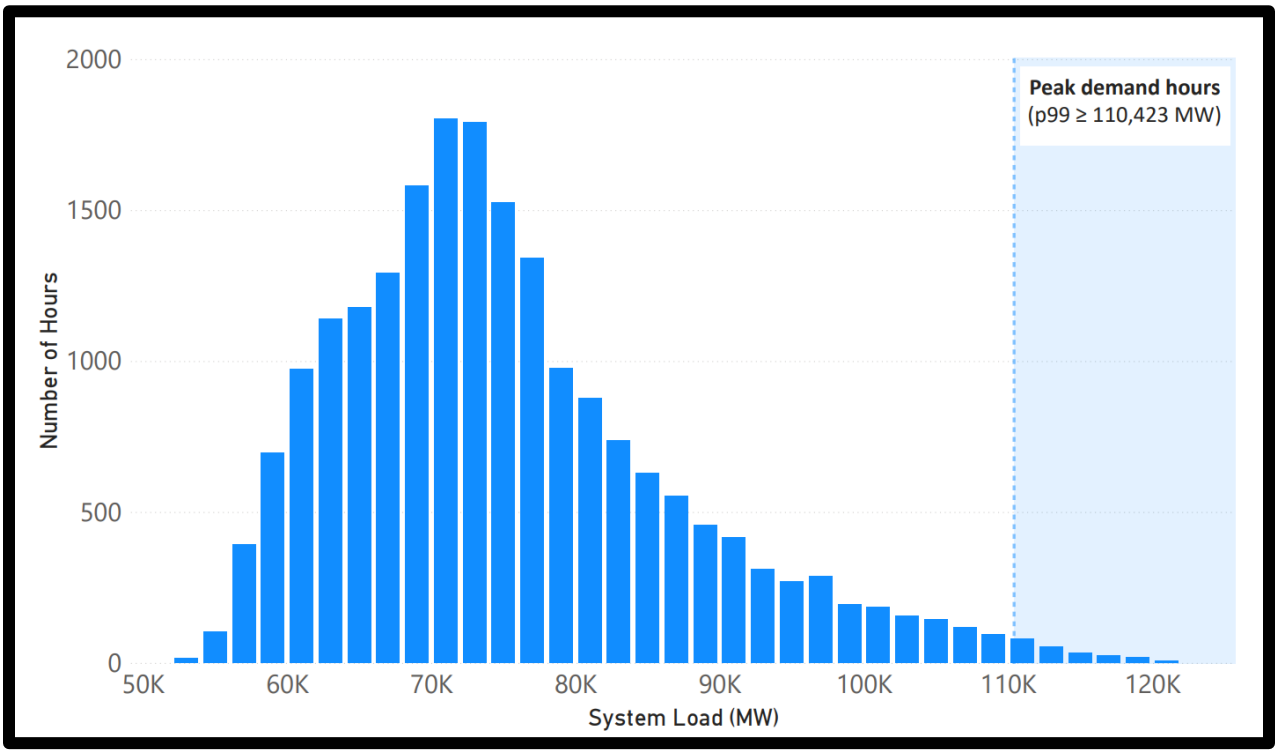

Figure 1.2. Distribution of Hourly System Load
The figure uses fixed-width (2,000-megawatt) bins (2023-09-01 to 2025-12-31). The dashed vertical line indicates the 99th percentile (p99) threshold used to define peak demand. Shaded bars represent hours meeting or exceeding this threshold.

Figure 1.2 visualizes the full distribution of hourly system load using fixed-width megawatt bins. The distribution is unimodal, with most observations concentrated well below the upper tail, and exhibits a pronounced right skew as load increases. The dashed vertical line marks the 99th-percentile threshold used throughout this study to define peak demand (**110,423 MW**), while the shaded region highlights hours at or above this level.

The visual separation between the bulk of observations and the shaded peak region is striking. System load observations cluster tightly around typical operating levels, while the frequency of hours declines rapidly as load approaches the upper tail. Beyond the p99 threshold,

observations become sparse, indicating that peak demand conditions do not arise as a smooth continuation of normal system behavior but instead represent a distinct and infrequent operating regime.

By definition, peak demand hours are rare. Across 20,472 observed hourly intervals in the study window, **only 205 hours—approximately 1.0% of all observations—meet or exceed the p99 threshold**. These hours define the system's most extreme demand states and are operationally significant despite their infrequency.

Having shown that peak demand hours occupy a small, well-defined upper tail of the system load distribution, the analysis now turns from magnitude to timing. Understanding when these extreme load conditions occur is essential for operational planning and resource adequacy, as temporal clustering can amplify system stress even when individual events are infrequent.

How are peak demand hours distributed across years?

Using the 205 peak demand hours identified in the previous section, this subsection examines their temporal distribution, focusing on how the frequency of extreme load conditions changes across years.

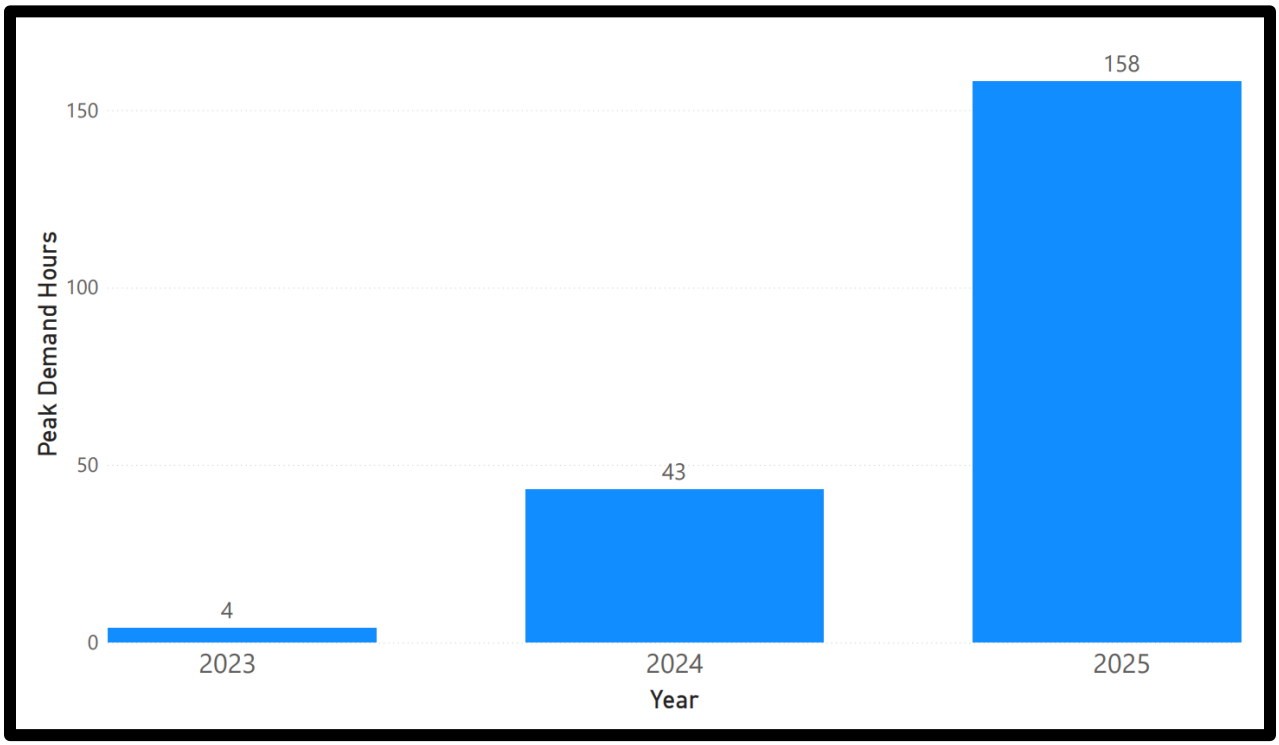

Figure 2.1. Annual Distribution of Peak Demand Hours (p99)
Peak demand hours are defined as the 205 hourly intervals with system load at or above the 99th percentile threshold.

Figure 2.1 shows a pronounced concentration of peak demand hours in the most recent year of the study period. Of the 205 hours meeting or exceeding the p99 threshold, **158 hours—approximately 77% of all peak demand observations—occur in 2025**. By comparison, 43 peak hours occur in 2024, while only 4 occur in 2023.

The limited contribution from 2023 reflects partial-year coverage, as the study window begins on September 1, 2023 and includes only four months of observations for that year. Even when accounting for this constraint, the progression from 2024 to 2025 reveals a substantial increase in the frequency of extreme load conditions. **Peak demand hours more than triple year over year,** indicating that high-load system states are occurring with markedly greater regularity in the most recent period.

This pattern is consistent with a system experiencing rising demand pressure rather than sporadic or idiosyncratic spikes. While this analysis does not attribute causality, the increasing concentration of peak demand hours in later years provides a clear signal that extreme demand conditions are becoming more frequent over time.

While Figure 2.1 captures the distribution of individual peak demand hours across years, it does not indicate how those hours cluster within the calendar. The next subsection reframes the analysis at the day level, counting the number of calendar days that contain at least one peak demand hour. This distinction is important for separating isolated extreme spikes from repeated entry into high-load operating states.

How are peak demand days distributed across years?

While the previous subsection examined the frequency of individual peak demand hours, this section shifts the focus to the calendar-day level. Counting the number of days that contain at least one peak demand hour distinguishes isolated extreme spikes from repeated entry into high-load operating states, which is more relevant for operational planning and system stress assessment.

Figure follows on the next page.

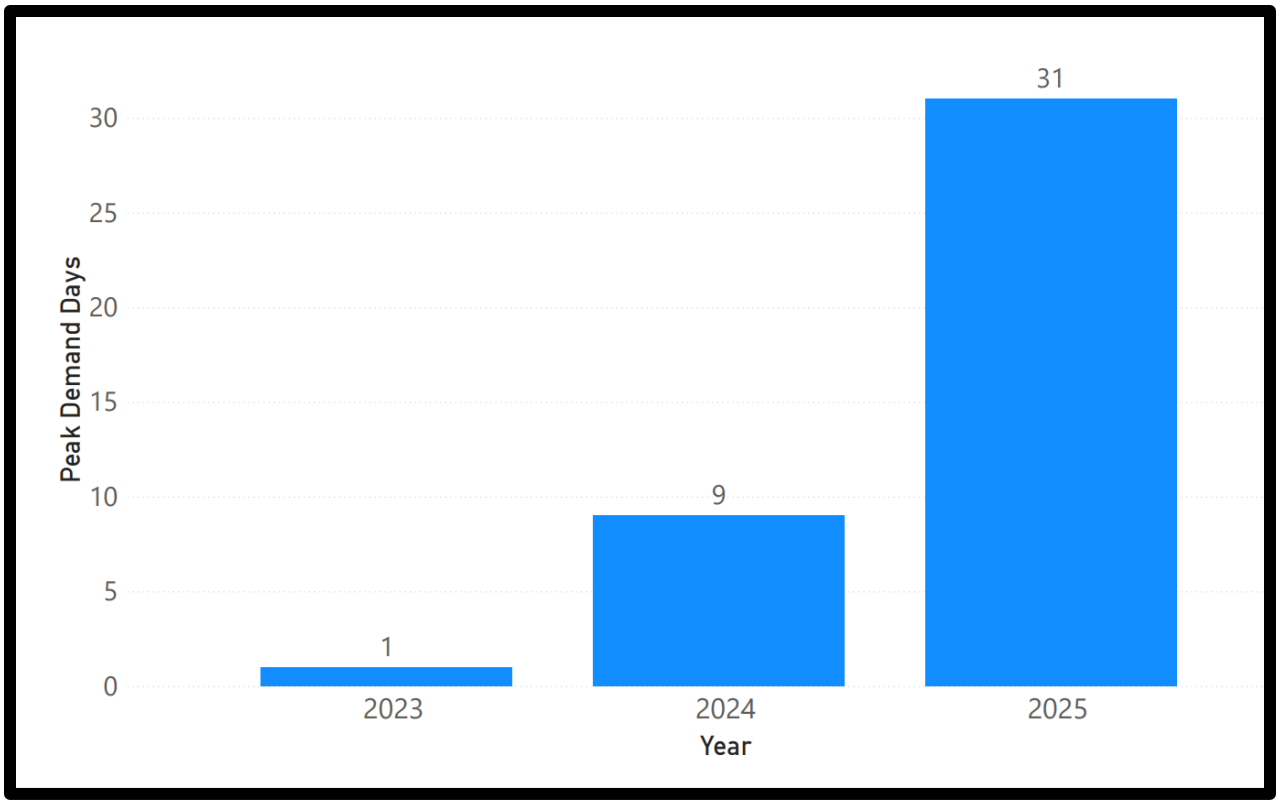

Figure 2.2. Number of Peak Demand Days by Year (2023–2025)
Peak demand days are defined as calendar days containing at least one of the 205 peak demand hours (≥ p99).

Figure 2.2 shows a clear increase in the number of days on which the system enters peak demand conditions. In 2023, all four peak demand hours occurred on a single calendar day, reflecting the limited temporal coverage of that partial year. In 2024, peak demand activity expanded to nine distinct days. By 2025, the system reached peak demand conditions on 31 separate days.

This represents an approximately **3.4-fold increase** in the number of peak demand days from 2024 to 2025. When viewed across the full study period, peak demand days are heavily concentrated in the most recent year: of the 41 total peak demand days observed between 2023 and 2025, roughly **75% occurred in 2025**, compared with **22% in 2024** and **less than 3% in 2023**. This pattern indicates a rapid escalation in the frequency with which the system reaches extreme load conditions.

Despite this growth in both the absolute number and relative concentration of peak demand days, the internal structure of peak events remains remarkably stable. Across years, peak demand days typically contain approximately four to five peak hours per day, suggesting that individual peak events have not materially lengthened or intensified.

This distinction is operationally important. The observed increase in peak exposure is driven by a sharp rise in the number of days experiencing extreme load conditions—particularly between 2024 and 2025—rather than by longer or more severe peak demand events within a given day. In other words, peak demand is becoming more frequent, not more prolonged.

Having determined how often peak demand occurs at the calendar-day level, the analysis next examines when during the year these peak demand days tend to occur, beginning with seasonal concentration.

Is peak demand seasonally concentrated?

This section examines the seasonal timing of peak demand conditions. The objective is to assess whether extreme system load is broadly distributed across the calendar year or concentrated within specific seasons.

Season	Peak Demand Hours	Peak Demand Days
Summer	201	40
Fall	4	1
Winter	0	0
Spring	0	0

Table 2.2. Seasonal Distribution of Peak Demand Hours and Days
Peak demand days are defined as calendar days containing at least one of the 205 peak demand hours.

Table 2.2 provides contextual evidence of the seasonal concentration of peak demand by summarizing both the number of peak demand days and the total number of peak demand hours by season. The results show that **peak demand in the MISO system is overwhelmingly a summer phenomenon**. Of the 205 peak demand hours identified, 201 occur during the summer months and are distributed across 40 summer days. Outside of summer, only a single peak demand day is observed, accounting for four peak hours in the fall. No peak demand hours occur during winter or spring under the top-205-hour definition used in this study.

This pattern demonstrates that summer dominance persists not only at the day level but also at the hourly level. While winter and spring may experience periods of elevated demand, they do not meaningfully contribute to the extreme upper tail of the load distribution captured in this analysis.

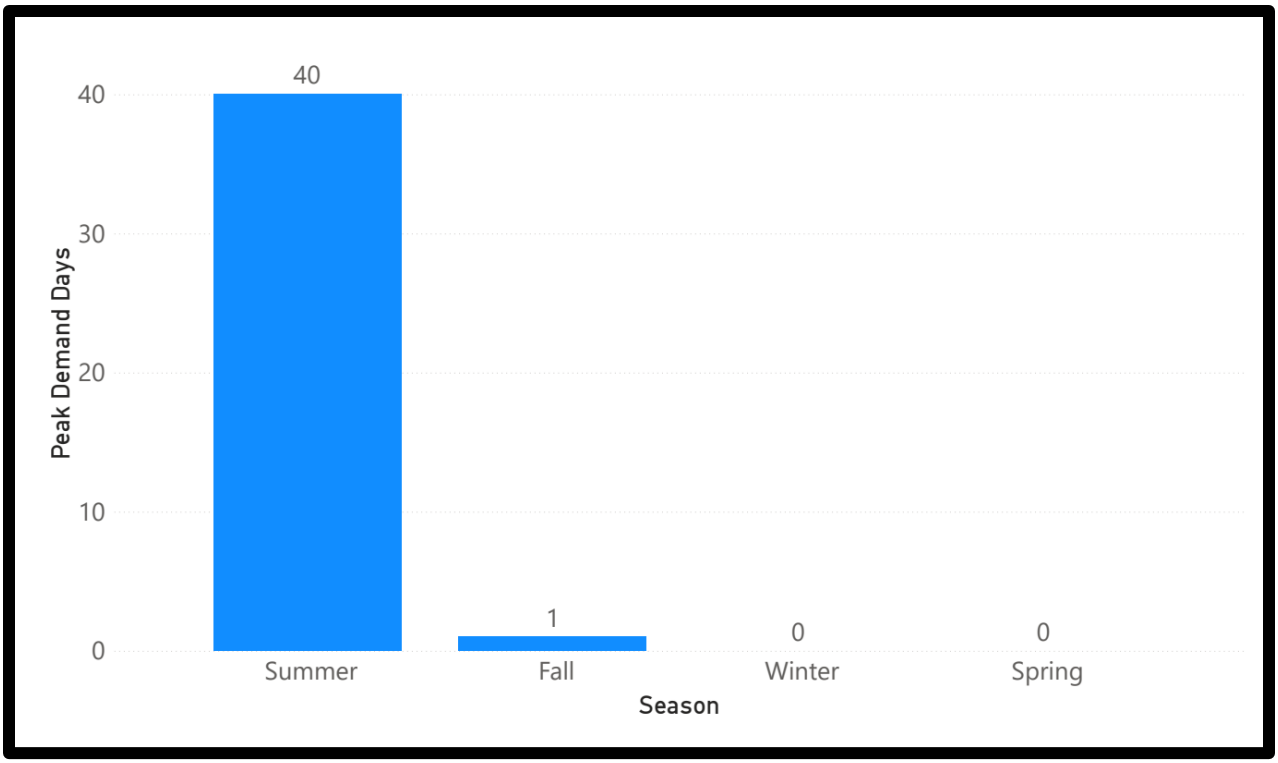

Figure 2.3. Number of Peak Demand Days by Season (2023-09-01 to 2025-12-31)

Figure 2.3 reinforces this seasonal concentration visually by illustrating the frequency of peak demand days by season. The figure highlights that **peak demand days occur almost exclusively during the summer**, reflecting repeated entry into high-load operating states within a relatively narrow portion of the calendar year.

Taken together, the table and figure establish that peak demand in the MISO system is highly seasonal, with extreme load conditions concentrated almost entirely in summer. Peak exposure is not evenly distributed across the year but instead clusters into a limited number of high-risk days during specific summer periods.

To refine this seasonal picture further, the next subsection examines how peak demand days are distributed by month. Moving from broad seasonal categories to a month-by-month view allows identification of when within the summer peak risk is most acute and whether extreme system load is confined to a narrow window or spread more evenly across the season.

What months do peak demand days occur?

Having shown that peak demand in the MISO system is overwhelmingly concentrated in summer, this subsection refines the seasonal view by examining how peak demand days are distributed within the summer months themselves. Moving from broad seasonal categories to a month-by-month perspective helps identify when within the summer peak risk is most acute.

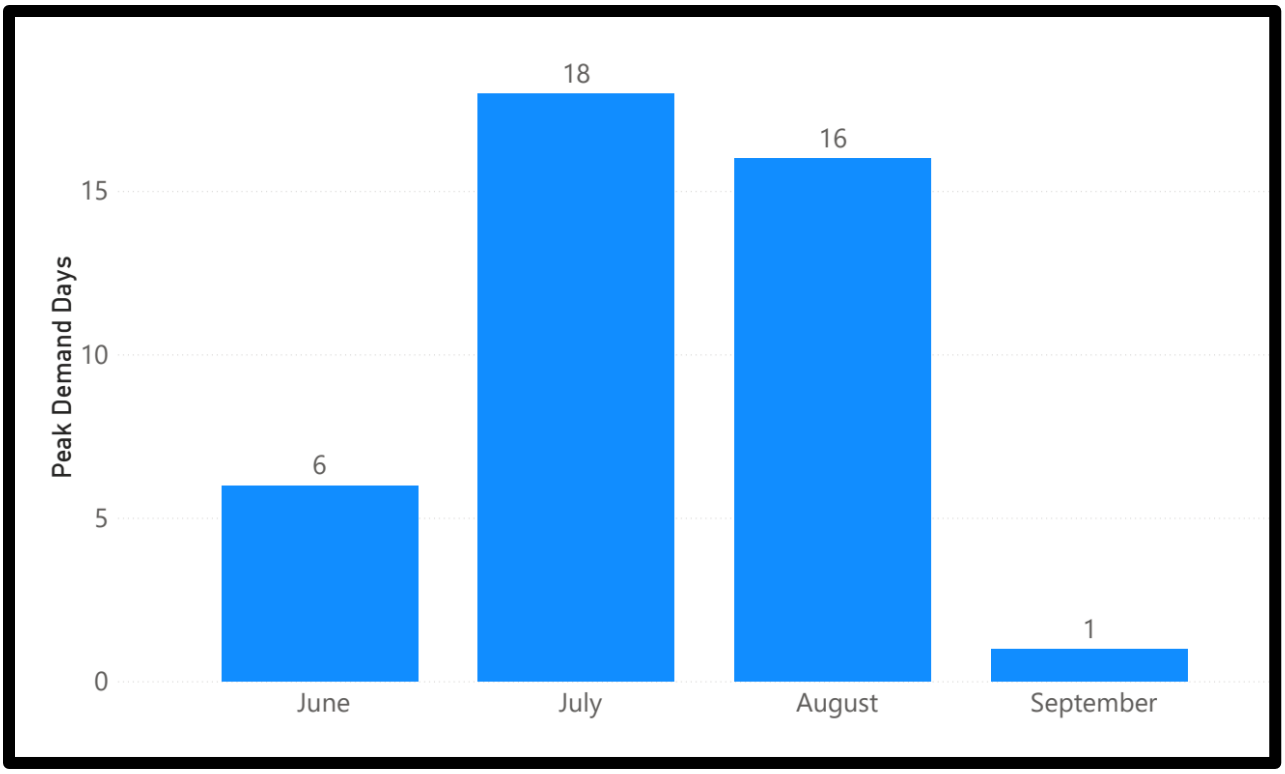

Figure 2.4. Monthly Distribution of Peak Demand Days
This figure presents the monthly distribution of peak demand days, defined as calendar days containing at least one of the 205 peak demand hours identified in Section 1.

Peak demand days are tightly concentrated in midsummer, with July and August accounting for the majority of occurrences. **July and August together account for approximately 83% of all peak demand days**, underscoring the strong seasonal concentration of extreme load conditions in midsummer. **June contributes an additional 15%, while September accounts for only 2%**. No peak demand days occur outside the June–September window.

This monthly pattern refines the seasonal findings by demonstrating that peak demand exposure is not evenly distributed across the summer period. Instead, extreme system load conditions are driven by a narrow set of high-stress months, with July and August emerging as the dominant contributors. The sharp drop-off outside these months underscores the temporal concentration of peak risk within a relatively brief portion of the calendar year.

After analyzing when peak demand days occur within the year, the analysis next examines how these days are distributed across the week, comparing weekday and weekend occurrence relative to the underlying calendar baseline.

Do peak demand days occur more often on weekdays or weekends?

Comparing weekday and weekend occurrence helps determine whether peak demand risk simply reflects the underlying calendar structure or is meaningfully associated with weekday system behavior.

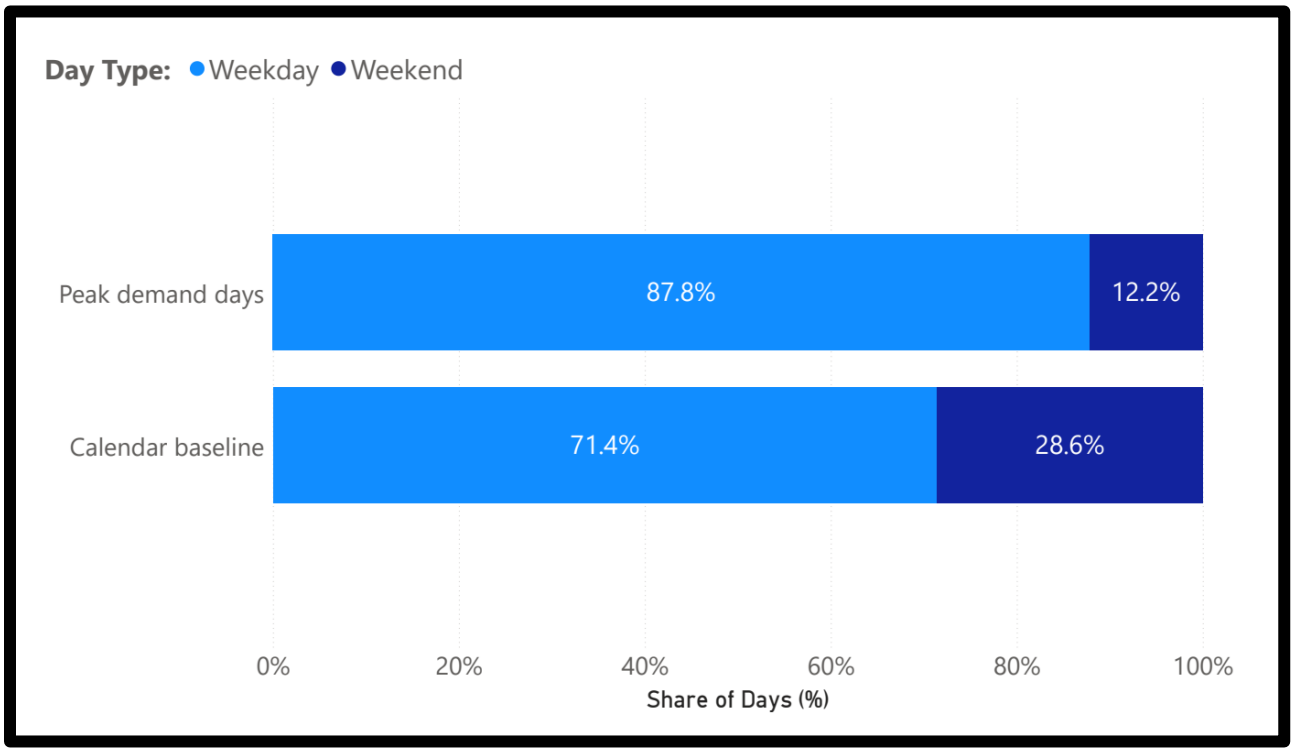

Figure 2.5. Weekday vs. Weekend Occurrence of Peak Demand Days
Comparison of peak demand day distribution with the expected calendar weekday–weekend baseline

Figure 2.5 compares the share of peak demand days occurring on weekdays and weekends against a calendar baseline that reflects the expected weekday–weekend split in a typical week. By default, approximately 71% of calendar days are weekdays and 29% are weekends; this baseline provides a neutral reference for evaluating whether peak demand days occur disproportionately on certain day types.

Peak demand days deviate substantially from this baseline. Of the days containing at least one peak demand hour, 87.8% occur on weekdays, while only 12.2% occur on weekends. Relative to calendar availability, peak demand days are approximately **1.23 times more likely** to occur on weekdays than expected—equivalent to a **23% higher likelihood** compared with a neutral calendar distribution.

This elevated weekday concentration indicates that peak demand risk is not evenly distributed across the week and cannot be explained by calendar structure alone. Instead, peak conditions are closely associated with weekday system behavior, consistent with higher levels of commercial, industrial, and institutional load that are less prevalent on weekends. While this analysis does not isolate specific demand drivers, it demonstrates that peak demand is systematically more likely to emerge during standard business days.

Having established that peak demand days are seasonally concentrated, clustered within specific months, and more likely to occur on weekdays, the analysis next examines intra-day timing to identify when, within a given day, peak demand hours most often occur.

At what times of day do peak demand hours most often occur?

This section examines the intraday timing of peak demand to determine whether extreme system load is evenly distributed across the operating day or concentrated within specific hours. By focusing on the 205 peak demand hours identified in Section 1, the analysis isolates periods of highest system load and evaluates how they align with daily load cycles.

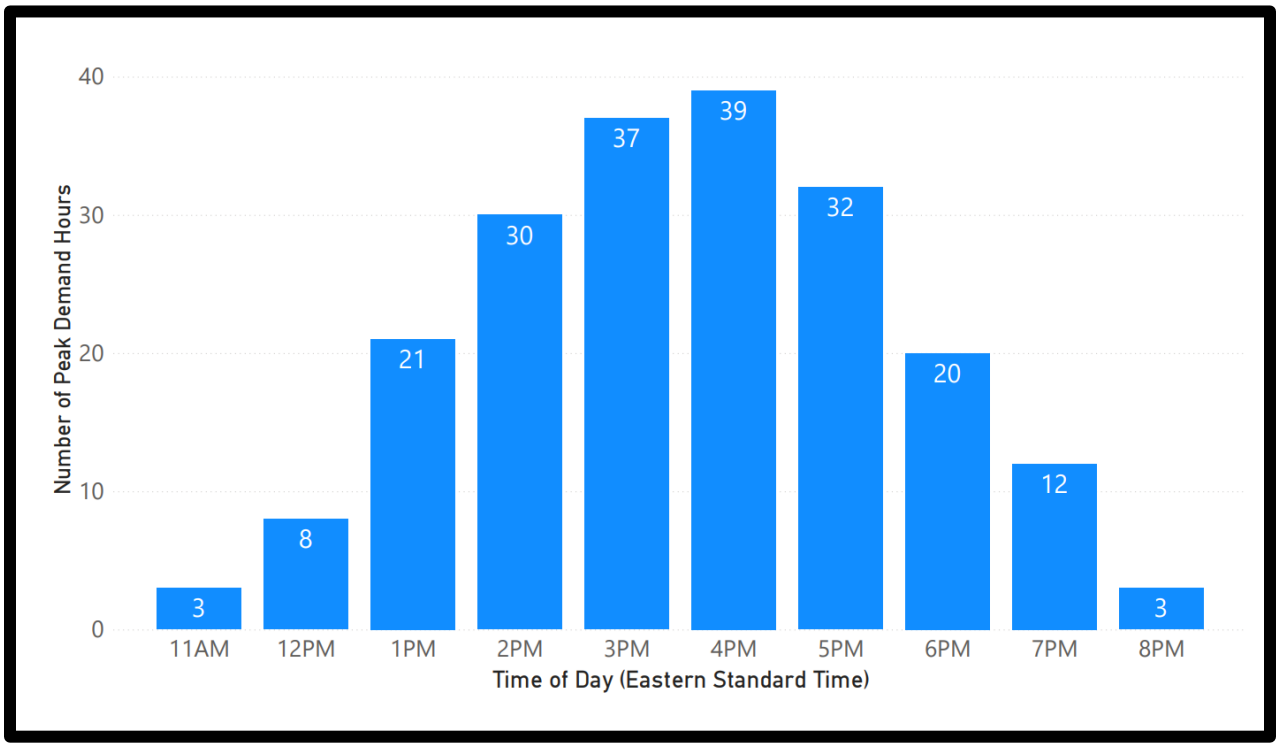

Figure 2.6. Distribution of Peak Demand Hours by Time of Day
Peak demand hours (≥ p99) grouped by clock time in Eastern Standard Time.

Figure 2.6 reveals a pronounced intraday concentration of peak demand hours in the afternoon and early evening. Peak activity begins to rise sharply after midday, accelerates through the early afternoon, and reaches its highest frequency between 3:00 PM and 5:00 PM EST, before tapering off into the evening.

Across the full distribution, approximately **87% of all peak demand hours occur between 1:00 PM and 6:59 PM EST**. This six-hour window captures the vast majority of extreme load conditions observed in the dataset, indicating that peak system load is tightly linked to afternoon and early-evening demand dynamics rather than being spread evenly across the day.

Peak demand hours are rare outside this window. Morning hours contribute minimally, and occurrences decline rapidly after 7:00 PM, suggesting that overnight and early-day usage patterns do not meaningfully contribute to extreme system load.

Taken together, this intraday profile indicates that peak demand emerges from a convergence of afternoon load drivers—such as sustained commercial activity combined with rising residential cooling demand—rather than from isolated or randomly timed events. With this temporal pattern established, the analysis next examines how peak demand hours cluster into multi-hour events and how long elevated demand conditions persist once they emerge.

How long do peak demand events persist?

This section examines the persistence of peak demand once extreme system load conditions emerge. Rather than treating peak demand as isolated hourly occurrences, the analysis groups consecutive peak demand hours into discrete events, allowing peak conditions to be evaluated as sustained operating states rather than momentary excursions.

Figure follows on the next page.

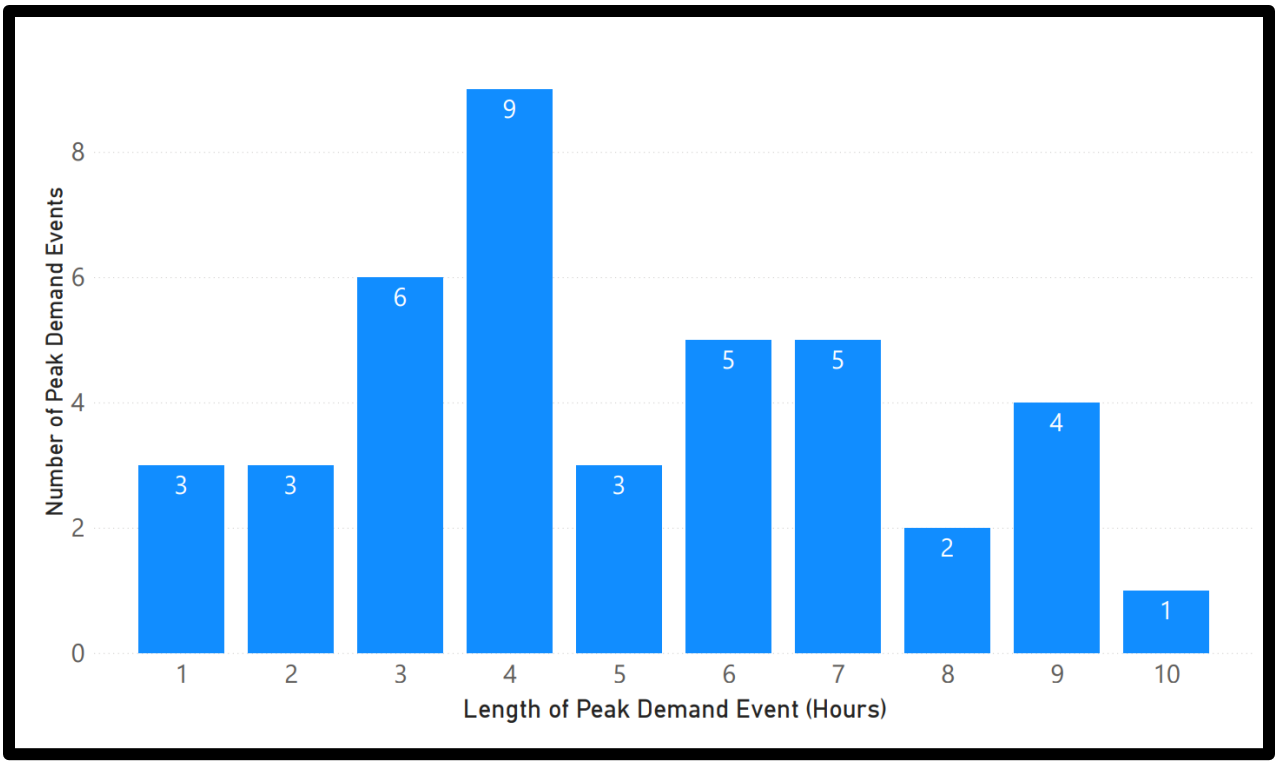

Figure 3.1. Distribution of Peak Demand Event Durations
Number of peak demand events by event length, measured as consecutive hours at or above the p99 system load threshold.

Figure 3.1 shows that peak demand events in MISO exhibit a clear multi-hour structure. Event durations range from 1 to 10 consecutive hours, with a pronounced concentration between **3 and 7 hours, which together account for approximately 68% of all peak demand events**. The most common duration is **four hours**, which occurs more frequently than any other event length.

Single-hour peak demand events are relatively uncommon, indicating that extreme system load rarely appears as a brief spike. At the same time, while longer events extending across much of the day do occur, they are progressively less frequent. Notably, no peak demand events persist beyond ten hours, suggesting that even under extreme conditions, **elevated system load remains bounded within a single operating day.**

Across all observed peak demand events, **the median event duration is four hours**, meaning that half of all peak episodes persist for four consecutive hours or less. This median aligns closely with the modal event length, reinforcing the conclusion that peak demand in MISO is best characterized as a sustained, multi-hour operating condition rather than a short-lived anomaly.

Taken together, these results indicate that peak demand persistence reflects structured periods of elevated system load rather than isolated volatility. To further clarify the operational profile of these events, the next section examines how peak demand events are distributed across specific duration ranges, distinguishing between shorter, moderate, and longer-lasting peak conditions.

What duration ranges do peak demand events most commonly fall into?

This subsection refines the event-level persistence analysis by grouping peak demand episodes into operationally meaningful duration ranges. While the previous section established that peak demand typically persists for multiple consecutive hours, this view clarifies *how those hours distribute across shorter and longer event windows*.

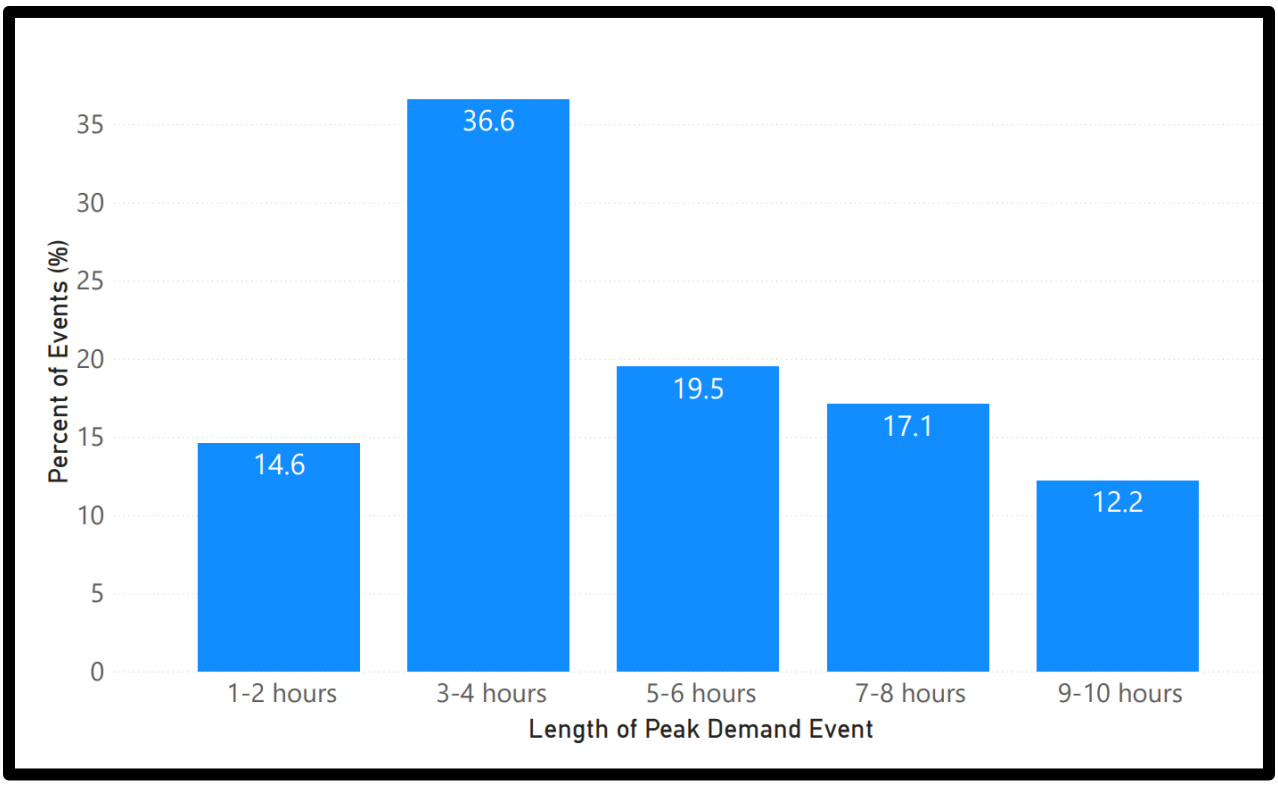

Figure 3.2. Peak Demand Events by Duration Range
Distribution of peak demand events grouped into two-hour duration buckets, shown as a percentage of total events.

Figure 3.2 shows that peak demand events most commonly fall into the 3–4 hour range, which accounts for **36.6% of all observed events**. This bucket aligns directly with the median event duration identified earlier, reinforcing that a four-hour peak is a representative and recurring system condition rather than an edge case.

Beyond this central tendency, a substantial share of peak demand events persist longer. **Nearly 49% of events last five hours or more**, while approximately **29% extend seven hours or longer**. Although the longest events—those lasting 9–10 hours—are less frequent, they remain a meaningful portion of the distribution at 12.2%, indicating that prolonged system load is not uncommon during extreme demand periods.

Taken together, these duration ranges confirm that peak demand in MISO is best understood as a sustained operational state, not a brief excursion. Most peak events require multiple consecutive hours of elevated system performance, with a nontrivial share extending across much of the operating day.

With the persistence and duration of peak demand events established, the analysis next turns to *how these events form*, examining the system load conditions that typically precede entry into peak demand. This perspective helps distinguish gradual escalation from abrupt transitions into extreme load states.

What load conditions precede peak demand events?

A natural hypothesis is that peak demand events are preceded by unusually large short-term increases in system load—particularly sharp "ramp-up" behavior during the morning hours leading into the afternoon peak. If this were the case, peak demand days would be expected to exhibit systematically larger positive hour-to-hour load changes than non-peak days prior to reaching extreme load conditions.

To evaluate this hypothesis, exploratory analysis examined hour-to-hour changes in system load across peak demand days and non-peak days, with particular attention to positive load deltas during the morning period. While some peak demand days do experience notable ramps, average and median ramp magnitudes are broadly similar across the two groups. Moreover, the largest single-hour load increase observed in the dataset occurs on a non-peak day. Taken together, these findings indicate that sharp short-term ramps are **neither necessary nor sufficient** conditions for peak demand outcomes.

Because ramp intensity alone does not reliably distinguish peak demand days, the analysis shifts focus from *how fast load increases* to *the level at which load is already operating* prior to the afternoon peak window.

Figure follows on the next page.

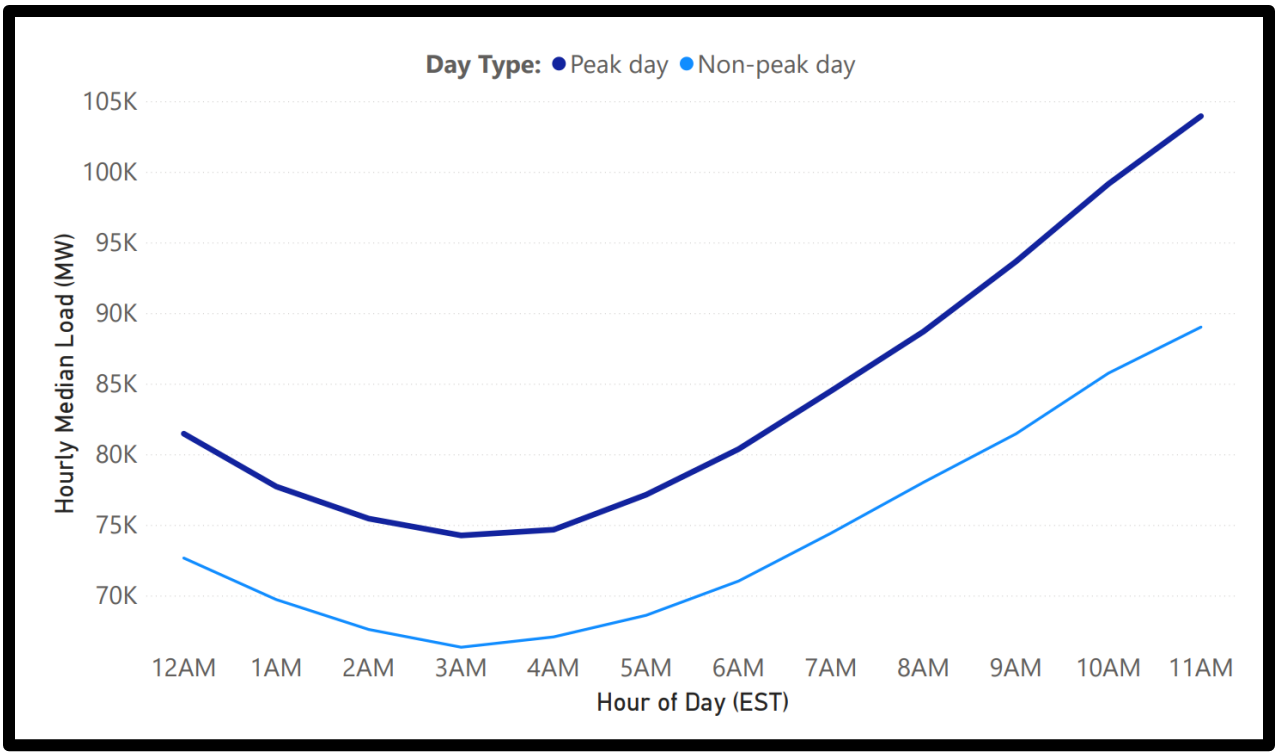

Figure 4.1. Morning System Load on Peak Demand Days vs. Non-Peak Days
Hourly median system load during overnight and morning hours (12:00 AM–11:00 AM EST), restricted to summer months. Peak demand days are defined as calendar days containing at least one p99 load hour.

Figure 4.1 compares hourly median system load profiles for peak demand days and non-peak days during the overnight and morning period, restricted to summer months to control for seasonality. Rather than diverging sharply through ramp behavior, the two profiles exhibit broadly parallel shapes, indicating similar intraday evolution at different absolute load levels.

Across the full window from midnight through late morning, median system load on peak demand days is consistently higher than on non-peak days by **approximately 8,000–10,000 MW at every hour**. This elevation persists throughout the morning and is not confined to a narrow pre-peak interval. By late morning, just before noon EST, **median system load reaches approximately 104,000 MW on peak demand days, compared with roughly 89,000 MW on non-peak days.**

These results reinforce the conclusion suggested by the ramp analysis: peak demand conditions do not arise from anomalous short-term accelerations in load, but from persistently elevated system demand that is already in place well before the afternoon peak window. Peak demand afternoons are therefore better understood as the culmination of sustained system pressure rather than the outcome of sudden intra-day dynamics.

In summary, while short-term ramps were explicitly examined and found to be inconclusive as drivers of peak demand, the underlying level of system load emerges as the dominant precursor to extreme demand conditions. This finding shifts the analytical emphasis away from intra-day volatility and toward the structural and spatial composition of demand across the system.

With the temporal preconditions for peak demand established, the analysis now turns to regional load behavior, examining how different parts of the MISO footprint contribute to sustained system-level demand and whether regional composition shifts meaningfully during peak demand periods.

How do MISO regions differ in demand?

Before analyzing how individual regions contribute during peak demand conditions, it is first necessary to establish baseline differences in regional load across the MISO footprint. Understanding the typical scale and distribution of demand in each region provides essential context for interpreting regional behavior under extreme system load.

Table 5.1 summarizes key distributional statistics for hourly load by region over the full study period, including minimum, average, median (50th percentile), 99th percentile, and maximum observed load. These metrics characterize both normal operating conditions and upper-tail behavior, which is particularly relevant for peak demand analysis.

Region	Minimum	Average	Median	99th Percentile	Maximum
Central	25,525	37,728	36,754	57,904	65,778
South	13,526	20,046	19,134	30,540	34,401
North	12,524	17,451	17,214	23,754	26,639

Table 5.1. Regional Hourly Load Summary Statistics (MW)
Summary statistics computed across all hours in the analysis period.

As shown in Table 5.1, the Central region carries the largest share of system load across all metrics. Its median hourly load is **approximately 36,754 MW—nearly double that of the South (19,134 MW) and North (17,214 MW) regions**. This ordering persists at the upper tail of the distribution: the Central region's 99th-percentile load reaches 57,904 MW, compared to 30,540 MW in the South and 23,754 MW in the North.

To visualize these distributional differences more clearly, Figure 5.1 compares the median and 99th-percentile load for each region. This paired representation highlights both typical operating levels and the magnitude of extreme load conditions within each region.

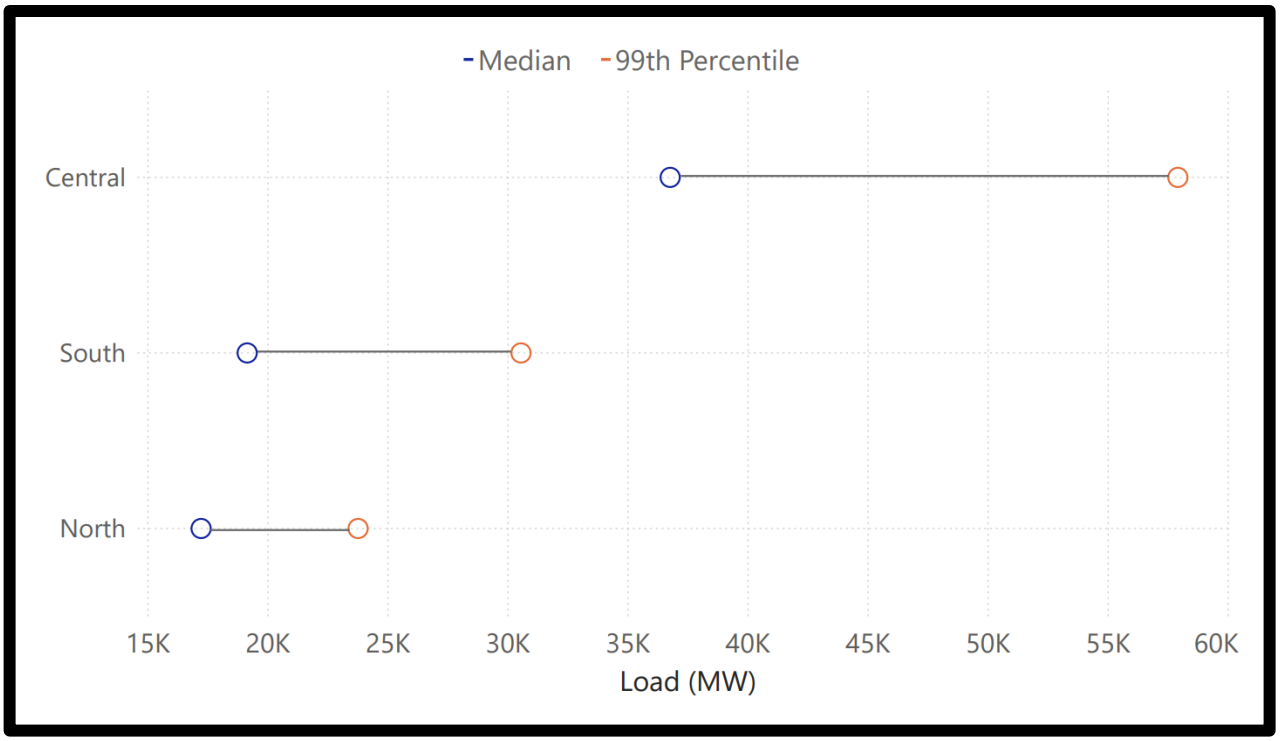

Figure 5.1. Regional Median and 99th-Percentile Hourly Load (MW)
Points represent the median (50th percentile) and 99th-percentile hourly load for each region over the full analysis period.

Figure 5.1 reinforces the Central region's dominant role in absolute load contribution across the MISO system. At both the median and 99th percentile, Central load levels substantially exceed those of the South and North regions, reflecting its larger customer base and higher baseline demand. From a system planning perspective, this establishes the Central region as the primary driver of aggregate system load.

At the same time, the relative spacing between median and upper-tail load is broadly similar across all three regions. While absolute magnitudes differ, each region exhibits a comparable proportional increase from typical conditions to extreme load hours. This pattern suggests that regional demand scales upward in parallel during high-load conditions rather than diverging sharply.

This distinction is important: while Table 5.1 and Figure 5.1 clarify where system load is concentrated, they do not indicate whether peak demand conditions materially alter the regional composition of the system. The next section addresses this directly by examining whether regional load shares shift during peak demand hours, or whether peak demand reflects a broadly synchronized increase across the MISO footprint.

Is peak demand driven by a shift in regional load shares?

With baseline regional demand levels established, the next question is whether peak demand conditions materially alter the composition of system load across the MISO footprint. If peak demand were driven by a particular region disproportionately surging, this would appear as a meaningful shift in regional load shares during peak hours relative to typical system operation.

Figure 5.2 addresses this question by comparing the median regional share of total system load during peak demand hours to that observed during non-peak hours. Shares are computed as each region's median hourly contribution to total system load, providing a stable representation of system composition under different demand conditions.

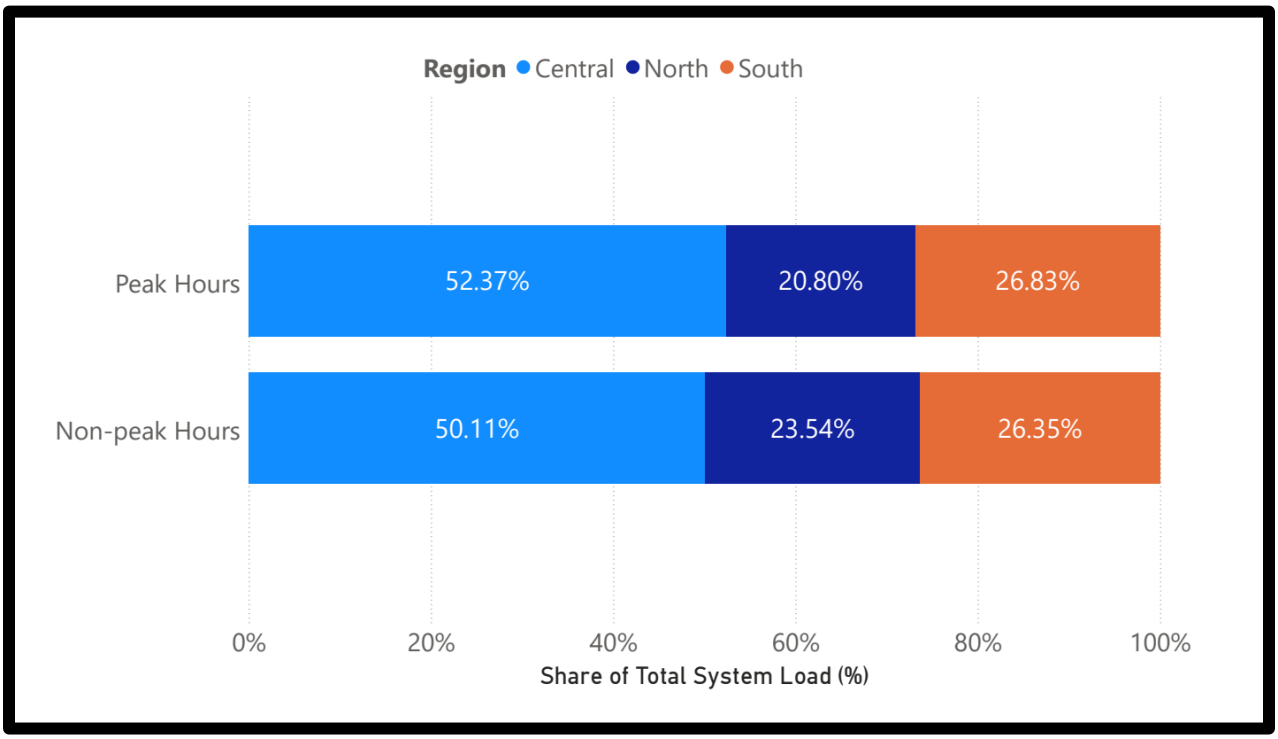

Figure 5.2. Regional Share of Total System Load During Peak and Non-Peak Hours
Bars show the median share of total system load contributed by each region during peak demand hours (≥ p99) and non-peak hours.

The results indicate a high degree of structural stability. During non-peak hours, the Central region accounts **for approximately 50.1% of total system load, followed by the South at 26.4% and the North at 23.5%**. During peak hours, the Central region's share increases modestly to 52.4%—**a gain of 2.3 percentage points**—while the South remains nearly unchanged at 26.8%. The North correspondingly yields share, declining to 20.8%, **a reduction of 2.7 percentage points.**

This pattern reflects a modest reweighting rather than a structural shift. No region exhibits a disproportionate or anomalous response to peak demand conditions. Instead, peak demand appears to arise from broadly synchronized increases across the system, with the Central region absorbing a slightly larger share of incremental load while the South remains stable and the North contributes relatively less.

The absence of dramatic regional divergence reinforces the conclusion that peak demand in MISO is not driven by a single regional outlier, **but rather by system-wide scaling of demand.** This consistency supports both operational planning and longer-term resource adequacy assessments, as it suggests that peak load reflects aggregate conditions rather than localized regional shocks.

To assess whether this regional stability persists throughout the year—or whether seasonal dynamics introduce more pronounced regional differences—the next section examines how regional load varies across seasons and how those patterns interact with peak demand risk.

How does seasonal load behavior differ across MISO regions?

With regional load scale and peak-hour composition established, the final step is to examine how demand evolves across seasons within each MISO region. Seasonal patterns provide important context for assessing whether peak demand risk reflects short-term operating anomalies or recurring structural dynamics embedded in regional demand profiles.

To isolate these effects, median hourly load is examined by season for each region. The use of medians ensures that comparisons reflect typical seasonal behavior rather than being driven by a small number of extreme hours.

Figure follows on the next page.

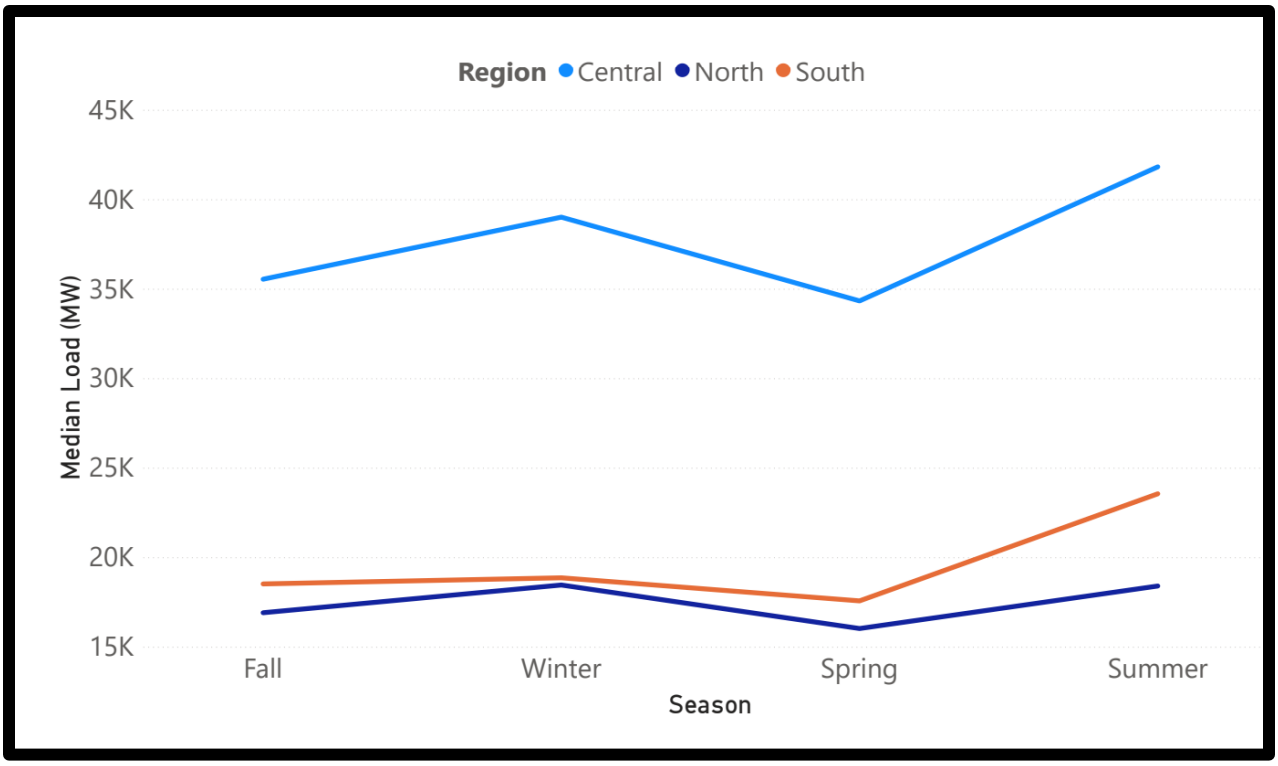

Figure 5.3. Median Hourly Load by Season and Region
Lines show the median hourly load within each season for the Central, North, and South regions across the full analysis period. Seasons are ordered chronologically (Fall, Winter, Spring, Summer).

Figure 5.3 reveals a clear and consistent seasonal structure across the MISO footprint. All three regions exhibit the same ordering: load rises into winter, dips into a trough in spring, and peaks in summer. This shared pattern indicates that system-wide seasonal drivers—such as temperature, daylight, and economic activity—shape demand behavior across the entire network.

While the seasonal sequence is consistent, the magnitude and sensitivity of response differ by region. **The Central and North regions display a more distinct winter–spring–summer arc, with elevated winter loads and a pronounced spring trough preceding the summer peak.** In contrast, the **South region remains comparatively flat from fall through spring, followed by a sharper upward inflection into summer.**

This divergence suggests that summer conditions play a more dominant role in shaping load behavior in the South, whereas the Central and North regions experience broader seasonal variation throughout the year. Importantly, no region exhibits anomalous or counter-cyclical behavior; regional differences reflect variation in degree rather than fundamentally different demand dynamics.

Taken together, these findings reinforce the conclusion that peak demand risk in MISO is a coordinated system-wide phenomenon. Regional loads move in parallel across seasons, with differences arising from sensitivity and scale rather than direction. The Central region consistently carries the largest absolute load, but neither peak conditions nor seasonal shifts introduce structural changes in regional composition. Instead, peak demand emerges from shared seasonal pressures acting across the footprint, with predictable and stable regional responses.

Summary of Findings

The findings below synthesize system-level, temporal, and regional analyses of observed load behavior across the MISO footprint. Together, these findings describe how peak demand manifests, when it occurs, and why its frequency has increased.

- **Peak demand in MISO reflects a recurring and structurally patterned operating condition rather than isolated or anomalous events.** System-level load increased over the study period, and peak demand hours and days became substantially more frequent in the most recent year. This shift reflects more frequent entry into extreme load conditions, not longer or more persistent individual peak events.
- **Peak demand follows a highly consistent temporal structure.** Extreme load hours occur almost exclusively during the summer months, disproportionately on weekdays, and are tightly clustered in the afternoon and early evening. Peak demand is therefore not evenly distributed across the calendar but emerges repeatedly within well-defined seasonal and intraday operating windows.
- **Peak demand events are typically multi-hour but bounded in duration.** Most events consist of several consecutive hours of elevated load rather than extended or multi-day episodes, indicating stable internal structure rather than escalating event severity.
- **Peak demand is not driven by sharp short-term load ramps.** Conditions leading into peak demand show that peak days are characterized by persistently elevated baseline load well before the afternoon peak window. Morning load levels on peak days are consistently higher than on non-peak days, with parallel hourly trajectories rather than distinct ramp signatures.
- **Peak demand is a system-wide phenomenon rather than a regional outlier.** While the Central region carries the largest share of system load in absolute terms, regional load shares remain broadly stable during peak demand hours, with only modest and predictable reweighting relative to non-peak conditions.
- **Seasonal load behavior is broadly consistent across regions.** All regions exhibit a summer-dominant load profile, with no evidence of counter-cyclical or anomalous regional dynamics during extreme demand periods.
- **Overall, rising peak exposure reflects increasing baseline demand interacting with shared temporal and seasonal drivers.** The system is entering high-load operating states more frequently due to sustained demand growth, not because peak events themselves are becoming longer, more extreme, or more regionally concentrated.

Limitations & Next Steps

Limitations

This analysis is intentionally focused on observed system load behavior and is subject to several important limitations.

First, the study relies exclusively on demand-side data and **does not incorporate supply-side information.** Generation availability, unit commitment, outages, reserve margins, and transmission constraints are not considered. As a result, the analysis characterizes when and how high load occurs, but does not evaluate system stress, reliability risk, or adequacy relative to available resources.

Second, **no explicit weather data are included**. While temperature and humidity are known to be primary drivers of electrical demand—particularly during summer peak periods—this study does not attribute observed load patterns to specific meteorological conditions. Seasonal and temporal patterns are therefore described empirically rather than causally.

Third, the analysis **does not include market data** such as locational marginal prices (LMPs), congestion costs, or uplift payments. Consequently, it does not assess how load behavior translates into economic outcomes, market stress, or price volatility during peak demand conditions.

Finally, the study is **retrospective and descriptive in nature**. No forecasting, scenario analysis, or predictive modeling is performed. Observed patterns reflect historical system behavior over the study period and should not be interpreted as forecasts of future load levels or peak conditions.

These limitations are deliberate and reflect the study's focus on establishing a clear empirical baseline for system load behavior before introducing additional complexity.

Next Steps

Several natural extensions follow from the empirical patterns documented in this analysis.
A first extension would incorporate **weather data**, such as temperature, heat index, or cooling degree days, to quantify the relationship between meteorological conditions and both typical and extreme load behavior. This would enable attribution of peak demand events to specific weather drivers and improve interpretability of seasonal and intraday patterns.

A second extension would integrate **supply-side and operational data**, including generation capacity, outages, reserves, and transmission constraints. Coupling load behavior with system

conditions would allow evaluation of reliability margins, stress events, and the operational significance of increasingly frequent peak demand days.

A third and particularly compelling extension would examine **market outcomes** alongside load behavior. Incorporating real-time and day-ahead price data would enable analysis of how increasing load levels and peak demand frequency translate into price formation, volatility, and congestion patterns. Understanding whether price signals scale proportionally with load—or respond nonlinearly during extreme demand conditions—would provide valuable insight for market participants and planners alike.

Finally, future work could extend the analysis into **forecasting and scenario modeling**, using historical load, weather, and system data to evaluate how projected demand growth may affect peak frequency, duration, and regional contributions under different assumptions.

Together, these extensions would build on the descriptive foundation established here, moving progressively from characterization of load behavior toward causal inference, operational insight, and forward-looking system analysis.

Works Cited

[1] Federal Energy Regulatory Commission (FERC), "Summer Energy Market and Electric Reliability Assessment," 2024.

[2] C. Villarreal, "The Southern Value of Strengthening Competition on the Midcontinent Grid," R Street Institute, 6 December 2023. [Online]. Available: https://www.rstreet.org/commentary/southern-value-of-strengthening-competition-on-the-midcontinent-grid/. [Accessed January 2026].

[3] Midcontinent Independent System Operator (MISO), "MISO Fact Sheet," December 2025. [Online]. Available: https://cdn.misoenergy.org/Dec%202025%20Corp%20Fact%20Sheet732770.pdf. [Accessed January 2026].

[4] U.S. Energy Information Administration (EIA), "Results from the 2015 Residential Energy Consumption Survey (RECS): What's New in Home Energy Use?," U.S. Energy Information Administration, Washington, D.C., 2018.

[5] U.S. Energy Information Administration (EIA), "U.S. Data Center Energy Use," U.S. Energy Information Administration, Washington, D.C., 2024.

[6] U.S. Energy Information Administration (EIA), "Manufacturing Energy Consumption Survey (MECS)," U.S. Energy Information Administration, Washington, D.C., 2022.

[7] H. M. Hanna, "Energy Use for Field Operations, Crop Drying, and Swine Housing on University Farms," *ASABE,* 2016.

[8] Midcontinent Independent System Operator (MISO), "Actual Load (API documentation)," January 2026. [Online]. Available: https://data-exchange.misoenergy.org/api-details#api=load-generation-and-interchange-api&operation=get-v1-real-time-date-demand-actual. [Accessed January 2026].

[9] Midcontinent Independent System Operator (MISO), "2024 Reliability Imperative Report," Midcontinent Independent System Operator (MISO), Carmel, IN, 2024.